

CO2 Debugging

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QUESTIONS
Q1
1. Identify an error in the following snippet.
5 marks

Q2
2. Buggy code in code creation.
5 marks

Q3
3. Find the bugs, include the missing lines in the following snippet.
5 marks

3/3 answered

Q1 3. Identify an errors in the following snippet

5 marks

```
int binarySearch(int arr[], int n, int key)
{
    int low = 0, high = n - 1;
    while(low < high)
    { int mid = (low + high) / 2;
      if(arr[mid] == key) return mid;
      else if(arr[mid] < key)
        low = mid;
      else high = mid;
    }
    return -1;
}
```

Your Code

```
1 #include <stdio.h>
2 int binarySearch(int arr[], int n, int key)
3 {
4     int low = 0, high = n - 1;
5     while (low <= high)
6     {
7         int mid = (low + high) / 2;
8         if (arr[mid] == key)
9             return mid;
10        else if (arr[mid] < key)
11            low = mid + 1;
12        else
13            high = mid - 1;
14    }
15    return -1;
16 }
17
18 int main()
19 {
20     int n, key, i;
21     printf("Enter number of elements: \n");
22     scanf("%d", &n);
23     int arr[100];
24     printf("Enter %d sorted elements:\n", n);
25     for (i = 0; i < n; i++)
26         scanf("%d", &arr[i]);
27     printf("Enter key to search: \n");
28     scanf("%d", &key);
29     int result = binarySearch(arr, n, key);
30     if (result == -1)
31         printf("Element not found: \n");
32     else
33         printf("Element found at index %d\n", result);
34     return 0;
35 }
```

Submitted

Input

```
5
2 4 6 8 10
8
```

Output

```
Enter number of elements:
Enter 5 sorted elements:
Enter key to search:
Element found at index 3
```

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QUESTIONS

Q1 
1. Identify an error in the following snippet.
5 marks

Q2 
2. Buggy code in node creation.
5 marks

Q3 
3. Find the bugs, include the missing lines in the following snippet.
5 marks

3/3 answered

Q2 8. Buggy code in node creation

5 marks

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

int main() {
    struct Node *newNode;
    newNode->data = 10;
    newNode->next = NULL;
    printf("%d", newNode->data);
    return 0;
}
```

Your Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 struct Node {
4     int data;
5     struct Node *next;
6 };
7 int main() {
8     struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
9     if(newNode == NULL) {
10         printf("Memory not allocated");
11         return 0;
12     }
13     int x;
14     scanf("%d", &x);
15     newNode->data = x;
16     newNode->next = NULL;
17     printf("%d", newNode->data);
18     free(newNode);
19     return 0;
20 }
```

Input

10

Output

10

Submitted

QUESTIONS

Q1
3. Identify an array in the following snippet
5 marks

Q2
4. Debug code to make it work
5 marks

Q3
7. Find the bugs, include the missing lines in the following snippet
5 marks

3/3 answered

Q3 1. Find the bugs, include the missing lines in the following snippet.

5 marks

```
int linearSearch(int arr[], int n, int key) {  
    for(int i = 0; i <= n; i++) {  
        if(arr[i] == key)  
            return i;  
    }  
    return -1;  
}
```

Your Code

```
1 #include <stdio.h>  
2 int linearSearch(int arr[], int n, int key) {  
3     for (int i = 0; i < n; i++) {  
4         if (arr[i] == key) {  
5             return i;  
6         }  
7     }  
8     return -1;  
9 }  
10 int main() {  
11     int n, key;  
12     printf("Enter number of elements: \n");  
13     scanf("%d", &n);  
14     int arr[n];  
15     printf("Enter %d elements: \n", n);  
16     for (int i = 0; i < n; i++) {  
17         scanf("%d", &arr[i]);  
18     }  
19     printf("Enter key to search: \n");  
20     scanf("%d", &key);  
21     int index = linearSearch(arr, n, key);  
22     if (index != -1)  
23         printf("Key found at index %d\n", index);  
24     else  
25         printf("Key not found\n");  
26     return 0;  
27 }
```

Submitted

Input

```
5  
2 4 6 8 10  
8
```

Output

```
Enter number of elements:  
Enter 5 elements:  
Enter key to search:  
Key found at index 3
```